Crank–Nicolson Approach to the Two-Dimensional Schrödinger Equation

Solveig Juntunen, Tom Jussiaume, Erling Sandbekk
(Dated: December 11, 2025)

We simulate the time evolution of a two-dimensional Gaussian wave packet interacting with single-, double-, and triple-slit potentials by numerically solving the time-dependent Schrödinger equation. The Crank–Nicolson discretisation on a uniform grid yields an unconditionally stable scheme that preserves the norm of the wave function, with probability conserved to within 10^{-13}. The method accurately captures the expected behaviour of the packet, including reflection from the confining boundaries and transmission through the slit openings. By analysing the probability distribution on a detection screen, we quantify the diffraction and interference patterns generated by the different slit geometries, and the simulations reproduce the expected interference patterns for the single-, double-, and triple-slit configurations.

I. INTRODUCTION

The behavior of quantum systems is fundamentally governed by the time-dependent Schrödinger equation, whose solutions describe how probability distributions evolve in space and time [1]. In this project, we investigate this evolution for a single particle in two spatial dimensions by numerically solving the dimensionless Schrödinger equation on a unit-square domain with fixed boundary conditions. As a physical motivation, we study how a Gaussian wave packet interacts with barrier potentials designed to emulate classic slit experiments, where diffraction and interference arise from the wave-like nature of quantum particles.

To carry out this study, we implement a Crank–Nicolson discretisation of the equation and formulate the resulting update as a linear system involving large, sparse matrices. This approach allows us to simulate the propagation of the wave packet, verify numerical stability through probability conservation, and explore how different slit configurations influence the resulting interference patterns. [2].

The remainder of this report is structured as follows. In Sec. II we describe the numerical methods, including the discretisation strategy, construction of matrices, and setup of the initial state. Section III presents the results of the simulations, focusing on probability conservation, wave packet propagation, and interference patterns for various slit geometries. Section IV provides concluding remarks and outlines possible improvements for future work.

All code developed for this report is available via our GitHub repository.[1]

II. METHODS

The numerical methodology is built around four key components:

(i) formulating the governing equation on a unit-square domain with homogeneous Dirichlet boundary conditions,

(ii) discretising the spatial and temporal variables on a uniform Cartesian grid,

(iii) applying the Crank–Nicolson ($\theta = 1/2$) scheme to obtain a second-order accurate, unconditionally stable implicit time integrator [3], and

(iv) constructing the initial Gaussian wave packet and the time-independent barrier potentials that define the slit geometries.

All simulations employ a uniform $M \times M$ discretization of the spatial domain. The wave function u is complex-valued, and the potential v is real-valued and time-independent, leading to a sparse linear-system formulation for each Crank–Nicolson update.

The Schrödinger Equation

The evolution of a single, non-relativistic particle in two spatial dimensions is governed in position space by the dimensionless time-dependent Schrödinger equation

$$\mathrm{i}\,\frac{\partial u}{\partial t} = -\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} + v(x,y)\,u. \tag{1}$$

Here $u(x,y,t)$ is a complex-valued wave function and $v(x,y)$ is a real-valued, time-independent potential representing the slit barriers. We denote the imaginary unit by $\mathrm{i} = \sqrt{-1}$. All physical constants have been scaled away, simplifying the numerical implementation [4].

The probabilistic interpretation follows from the Born rule,

$$p(x,y;t) = |u(x,y,t)|^2 = u^*(x,y,t)u(x,y,t), \tag{2}$$

which represents the probability density of detecting the particle at (x,y) at time t [4]. The star in u^* denotes complex conjugation. The total probability

$$P(t) = \int_0^1 \int_0^1 |u(x,y,t)|^2 \,\mathrm{d}x\,\mathrm{d}y \tag{3}$$

[1] https://github.uio.no/erlinsa/fys3150-prj5

is normalized to unity and should remain equal to 1 for all t. We will use this normalization to monitor probability conservation in our simulations (Sec. III).

We impose homogeneous Dirichlet boundary conditions to confine the particle to the computational domain:

$$u(0,y,t) = u(1,y,t) = u(x,0,t) = u(x,1,t) = 0. \quad (4)$$

This choice simplifies the discretisation and assembly of the Crank–Nicolson scheme [5].

These boundary conditions effectively model a particle in a rigid "box" of side length 1. In particular, the wave packet is reflected at the domain boundaries rather than transmitted or absorbed, which also means that at sufficiently long times the dynamics are influenced by artificial reflections from the walls.

Discretisation of Space and Time

The spatial domain $[0,1]\times[0,1]$ is discretised on a uniform Cartesian grid with spacing h in both directions. We use M grid points in each direction and define

$$x_i = ih, \qquad y_j = jh, \qquad i,j = 0,\ldots,M-1,$$

so that $h = 1/(M-1)$. The wave function values are stored as

$$u^n_{ij} \equiv u(x_i, y_j, t_n),$$

and, since the potential $v(x,y)$ is time independent, it is sampled once on the same grid,

$$v_{ij} \equiv v(x_i, y_j).$$

The time interval $[0,T]$ is discretised uniformly as $t_n = n\Delta t$, where Δt is the time step and $n = 0,\ldots,N_t-1$ indexes the N_t time levels.

For interior indices $i,j = 1,\ldots,M-2$ on this uniform grid, we approximate the second derivatives using the standard second-order central differences [6],

$$\left.\frac{\partial^2 u}{\partial x^2}\right|_{ij} \approx \frac{u^n_{i+1,j} - 2u^n_{ij} + u^n_{i-1,j}}{h^2}, \quad (5)$$

$$\left.\frac{\partial^2 u}{\partial y^2}\right|_{ij} \approx \frac{u^n_{i,j+1} - 2u^n_{ij} + u^n_{i,j-1}}{h^2}, \quad (6)$$

which gives rise to the simple five-point 2D stencil for the discrete Laplacian [7],

$$\nabla^2_h u^n_{ij} = \frac{u^n_{i+1,j} + u^n_{i-1,j} + u^n_{i,j+1} + u^n_{i,j-1} - 4u^n_{ij}}{h^2}. \quad (7)$$

This spatial discretisation introduces an error of order $\mathcal{O}(h^2)$. When combined with the Crank–Nicolson time integrator (introduced in the next section), which is second-order accurate in time, the full scheme is second-order accurate in both space and time [6, 8].

Since all boundary values are fixed to zero for all time levels according to Eq. (4), only the interior points with $i,j = 1,\ldots,M-2$ are treated as unknowns. This reduces the number of unknowns from M^2 to $(M-2)^2$.

Crank–Nicolson Time Integration

To achieve numerical stability and second-order accuracy, we employ the Crank–Nicolson method. In the context of the Schrödinger equation, Crank–Nicolson is particularly well suited because its midpoint time discretisation yields an update operator that is approximately unitary to second order. As a result, the discrete evolution effectively preserves the norm of the wave function, mirroring the exact quantum mechanical requirement of probability conservation and avoiding artificial growth or decay of the total probability over long simulations [9, 10].

Conceptually, Crank–Nicolson can be viewed as a special case of the θ-method applied to the semi-discrete system,

$$\frac{u^{n+1} - u^n}{\Delta t} = \theta F(u^{n+1}) + (1-\theta)F(u^n),$$

where F denotes the spatial discretisation of the right-hand side of Eq. (1). Forward Euler corresponds to $\theta = 0$ (fully explicit), backward Euler to $\theta = 1$ (fully implicit), and Crank–Nicolson to $\theta = \frac{1}{2}$, which averages the two and yields a scheme that is unconditionally stable and second-order accurate in time [11].

Applying Crank–Nicolson to Eq. (1) yields, for each interior grid point,

$$\begin{aligned}
u^{n+1}_{ij} &- r\big(u^{n+1}_{i+1,j} - 2u^{n+1}_{ij} + u^{n+1}_{i-1,j}\big) \\
&- r\big(u^{n+1}_{i,j+1} - 2u^{n+1}_{ij} + u^{n+1}_{i,j-1}\big) + \frac{\mathrm{i}\Delta t}{2} v_{ij} u^{n+1}_{ij} \\
&= \\
u^n_{ij} &+ r\big(u^n_{i+1,j} - 2u^n_{ij} + u^n_{i-1,j}\big) \\
&+ r\big(u^n_{i,j+1} - 2u^n_{ij} + u^n_{i,j-1}\big) - \frac{\mathrm{i}\Delta t}{2} v_{ij} u^n_{ij},
\end{aligned} \quad (8)$$

where

$$r = \frac{\mathrm{i}\Delta t}{2h^2}. \quad (9)$$

For a full derivation of Eq. (8) see Appendix A.

To express this compactly, the interior values u^n_{ij} are packed into a vector $\vec{u}^{\,n}$ of length $K = (M-2)^2$. The vector $\vec{u}^{\,n}$, written here in row form for clarity, is constructed by stacking the interior grid values column-by-column,

$$\begin{aligned}
\vec{u}^{\,n} = \Big[&(u^n_{1,1}, u^n_{2,1}, \ldots, u^n_{M-2,1}), \\
&(u^n_{1,2}, u^n_{2,2}, \ldots, u^n_{M-2,2}), \\
&\ldots, \\
&(u^n_{1,M-2}, \ldots, u^n_{M-2,M-2})\Big].
\end{aligned}$$

Equivalently, we can describe the mapping from a pair of interior indices (i,j) (with $i,j = 1,\ldots,M-2$) to a single vector index k by

$$k = i + (M-2)(j-1), \qquad k = 1,\cdots,K,$$

so that $\vec{u}_k^n$ corresponds to u_{ij}^n under this ordering.

With this vectorisation, the Crank–Nicolson update can be written compactly as the linear system

$$A\,\vec{u}^{\,n+1} = B\,\vec{u}^{\,n}, \tag{10}$$

where $A, B \in \mathbb{C}^{K\times K}$ are sparse block-tridiagonal matrices. For the entry corresponding to grid point (i, j) with vector index k, the diagonal elements are

$$a_k = 1 + 4r + \frac{\mathrm{i}\Delta t}{2} v_{ij}, \qquad b_k = 1 - 4r - \frac{\mathrm{i}\Delta t}{2} v_{ij}.$$

The off-diagonal entries correspond to the couplings to the four nearest neighbours with weights $\mp r$ arising from the five-point stencil. Each time step consists of computing $\vec{b} = B\vec{u}^{\,n}$ and solving the linear system $A\vec{u}^{\,n+1} = \vec{b}$ using a sparse linear solver. Since A is constant in time for a time-independent potential v, its sparse LU factorisation can be computed once and reused in every time step [12].

Example of the Matrix Structure

To illustrate the sparsity pattern of the matrices A and B, consider the case $(M-2) = 3$, so that the interior grid contains $3 \times 3 = 9$ points. Using the row-by-row vectorisation described above, the matrices become 9×9 sparse block-tridiagonal matrices. Each interior point couples only to itself and its four nearest neighbours, resulting in at most five nonzero entries per row.

For $(M-2) = 3$, the matrix A takes the form

$$A = \begin{pmatrix} a_0 & -r & 0 & -r & 0 & 0 & 0 & 0 & 0 \\ -r & a_1 & -r & 0 & -r & 0 & 0 & 0 & 0 \\ 0 & -r & a_2 & 0 & 0 & -r & 0 & 0 & 0 \\ -r & 0 & 0 & a_3 & -r & 0 & -r & 0 & 0 \\ 0 & -r & 0 & -r & a_4 & -r & 0 & -r & 0 \\ 0 & 0 & -r & 0 & -r & a_5 & 0 & 0 & -r \\ 0 & 0 & 0 & -r & 0 & 0 & a_6 & -r & 0 \\ 0 & 0 & 0 & 0 & -r & 0 & -r & a_7 & -r \\ 0 & 0 & 0 & 0 & 0 & -r & 0 & -r & a_8 \end{pmatrix}.$$

Similarly, the matrix B has the same sparsity pattern but with r replacing $-r$ and b_k replacing a_k:

$$B = \begin{pmatrix} b_0 & r & 0 & r & 0 & 0 & 0 & 0 & 0 \\ r & b_1 & r & 0 & r & 0 & 0 & 0 & 0 \\ 0 & r & b_2 & 0 & 0 & r & 0 & 0 & 0 \\ r & 0 & 0 & b_3 & r & 0 & r & 0 & 0 \\ 0 & r & 0 & r & b_4 & r & 0 & r & 0 \\ 0 & 0 & r & 0 & r & b_5 & 0 & 0 & r \\ 0 & 0 & 0 & r & 0 & 0 & b_6 & r & 0 \\ 0 & 0 & 0 & 0 & r & 0 & r & b_7 & r \\ 0 & 0 & 0 & 0 & 0 & r & 0 & r & b_8 \end{pmatrix}.$$

These matrices illustrate the five-point stencil arising from the discrete Laplacian: each grid point couples only to its left, right, up, and down neighbours. For larger $(M-2)$ the same block-tridiagonal pattern repeats, producing matrices of size $(M-2)^2 \times (M-2)^2$ with the same local structure and overall sparsity.

Initial Wave Packet

The initial condition is a Gaussian wave packet $u(x, y, 0)$ equal to

$$\exp\left[-\frac{(x-x_c)^2}{2\sigma_x^2} - \frac{(y-y_c)^2}{2\sigma_y^2} + \mathrm{i}p_x x + \mathrm{i}p_y y\right], \tag{11}$$

evaluated on all interior grid points, with boundary values set to zero. Here, x_c and y_c denote the coordinates of the initial wave packet's center, σ_x and σ_y specify its widths along the x- and y-axes, and p_x and p_y represent its momenta. A Gaussian packet is a natural choice for this study: it is well-localised, analytically simple, and commonly used as a standard initial state in quantum wave-packet simulations.

After sampling the packet on the grid, the discrete wave function is normalised according to

$$\sum_{i,j} |u_{ij}^0|^2 = 1,$$

so that the discrete probability

$$p_{ij}^n = |u_{ij}^n|^2 \tag{12}$$

satisfies $\sum_{i,j} p_{ij}^0 = 1$.

Potential Construction

The potential is static and defines the geometry of the barrier and slits. It is implemented on the grid as

$$v_{ij} = \begin{cases} v_0, & \text{if } (x_i, y_j) \text{ lies inside a barrier region,} \\ 0, & \text{otherwise,} \end{cases}$$

with v_0 chosen sufficiently large to approximate impenetrable walls.

For the double-slit configuration, the barrier is a vertical wall centered at $x = 0.5$ with thickness 0.02 in x, i.e.

$$x \in [0.49,\ 0.51].$$

Two slit openings of width 0.05 are introduced symmetrically about $y = 0.5$, separated by a central wall segment of height 0.05. All grid points falling within the barrier rectangles are assigned $v_{ij} = v_0$, and all others are set to zero.

Time Evolution and Data Storage

Overall, a single simulation consists of (i) assembling the sparse matrices A and B from the chosen grid and potential, (ii) initializing and normalizing the wave packet, and (iii) performing a time loop in which the linear system (10) is solved repeatedly to advance the solution and compute derived quantities such as probabilities.

The discrete probability associated with grid cell (i, j) at time step n, given in Eq. (12), is interpreted directly as the probability of that grid cell, not as a continuous probability density.

The simulation evolves the wave function by iterating Eq. (10) from $n = 0$ to $n = N_t - 1$, yielding $N_t + 1$ time levels $u^0, \ldots, u^{N_t}$. At each step the solution is stored in memory, and later written directly to file for analysis.

The full time-stepping procedure is summarized in Algorithm 1.

Algorithm 1 Time evolution of the 2D Schrödinger equation using Crank–Nicolson

```
function SCHRODINGER_CN(M, h, Δt, N_t, V, u⃗^0)
    Construct sparse matrices A and B
        using M, h, Δt, and V
    Optionally precompute a sparse LU factorisation of A
    for n = 0 to N_t − 1 do
        b⃗ ← B u⃗^n
        Solve A u⃗^{n+1} = b⃗
        p_ij^{n+1} ← |u_ij^{n+1}|^2          ▷ probability per grid cell
```

Tools

All simulations were implemented in C++17 using the Armadillo 10.8.2 linear algebra library for complex-valued matrices and sparse matrix operations, compiled with GCC 11.4.0. The Crank–Nicolson system was solved at each time step using Armadillo's sparse solver `spsolve` with the `"superlu"` backend. Although the Crank–Nicolson matrix is time-independent and its sparse LU factorization could in principle be precomputed and reused, this functionality is not exposed in Armadillo 10.8.2. Implementing LU reuse would require either a newer Armadillo version or direct calls to the SuperLU API, and was therefore not pursued in this work.

Post-processing and figure generation were carried out in Python 3.10.12 using `matplotlib`, and animations were generated with `ffmpeg` version 4.4.2.

ChatGPT

ChatGPT was used as a supplementary assistant during the development of the C++, Python, and LaTeX components of the project. Its role was limited to tasks such as helping interpret compiler error messages, generating boilerplate structures, refining phrasing in the report and documenting the code. It was also consulted occasionally for clarification of conceptual questions or for comparing alternative implementation strategies.

All numerical methods, algorithmic choices, and final implementations were written, verified, and validated solely by the authors.

III. RESULTS AND DISCUSSION

Numerical Conservation of Probability

To assess probability conservation in the numerical solution of the two-dimensional time-dependent Schrödinger equation, we evaluated the total probability

$$P(t) = \sum_{i,j} p_{ij}(t)$$

at every time step for two simulations: (i) free propagation with no potential barrier ($v_0 = 0$), and (ii) propagation toward a double-slit barrier with a large potential height ($v_0 = 10^{10}$). We present the deviation from unity,

$$\Delta P(t) = P(t) - 1,$$

rather than $P(t)$ itself.

Figure 1 shows the time evolution of $|\Delta P(t)|$ over the simulation duration $T = 0.008$, using spatial resolution $h = 0.005$ and time step $\Delta t = 2.5 \times 10^{-5}$. In both the free-propagation and double-slit cases, the probability remains very close to unity: the deviation stays in the range $\sim 10^{-16}$–10^{-13} and does not exhibit any systematic growth over time. Within this parameter regime, we therefore observe good numerical conservation of probability for both configurations.

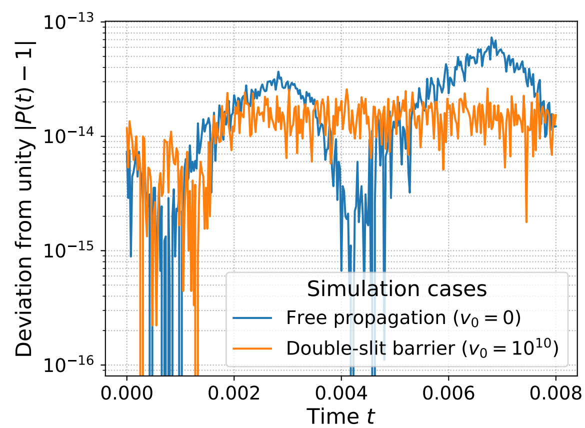

FIG. 1: Deviation of the total probability from unity, $|P(t) - 1|$, for both free propagation ($v_0 = 0$) and propagation through a double-slit potential barrier ($v_0 = 10^{10}$). A logarithmic scale is employed to resolve the extremely small deviations. Simulation parameters are listed in Table I.

We note that these tests were performed for a single choice of h, Δt and final time T. A more systematic study varying the grid spacing, time step and simulation horizon would be needed to fully map out the stability limits of the scheme and quantify how probability conservation degrades as the discretisation is coarsened.

For this study we employed the simulation parameters listed in Table I. All quantities are expressed in the dimensionless units of the scaled Schrödinger equation.

Symbol	Description	Value
h	Spatial grid spacing	0.005
Δt	Time step	2.5×10^{-5}
T	Final simulation time	0.008
M	Grid points in x, y	201
N_t	Number of time steps	321
x_c	Initial packet centre in x	0.25
y_c	Initial packet centre in y	0.50
σ_x	Initial width in x	0.05
σ_y	Initial width in y	0.05, $(0.10)^*$
p_x	Initial momentum in x	200
p_y	Initial momentum in y	0
v_0	Barrier height	0, $(10^{10})^*$

TABLE I: Simulation parameters for the probability–conservation study. Values marked with (*) correspond to the double-slit case; all others are identical to the free-propagation case.

Propagation of a 2D Gaussian Wave Packet Through a Double Slit

We simulated the time evolution of a two-dimensional Gaussian wave packet incident on a double-slit potential, using the numerical parameters listed in Table II. The wavefunction was evolved with the Crank–Nicolson scheme, and we extracted the probability distribution Eq (12) on the grid, as well as the real and imaginary parts of the wavefunction, at selected times.

Symbol	Description	Value
h	Spatial grid spacing	0.005
Δt	Time step	2.5×10^{-5}
T	Final simulation time	0.002
M	Grid points in x, y	201
N_t	Number of time steps	81
x_c	Initial packet centre in x	0.25
y_c	Initial packet centre in y	0.50
σ_x	Initial width in x	0.05
σ_y	Initial width in y	0.20
p_x	Initial momentum in x	200
p_y	Initial momentum in y	0
v_0	Barrier height (double slit)	10^{10}

TABLE II: Simulation parameters used in the double-slit experiment.

Figure 2 shows the evolution of the probability distribution $|u(x, y, t)|^2$ at $t = 0$, 0.001, and 0.002. At $t = 0$, the packet is well localised around its initial centre and exhibits the expected Gaussian envelope: narrow in the x-direction and significantly broader in the y-direction. By $t = 0.001$, the front of the packet has reached the barrier and only the parts aligned with the two slits have appreciable transmission, while the remainder is reflected by the high potential of the barrier. At the final time $t = 0.002$, the transmitted parts of the wave have propagated into the region to the right of the barrier and begin to form an interference pattern: the probability density displays alternating high- and low-intensity bands where the two transmitted regions overlap. The highest probability remains on the left side of the domain, indicating that most of the wave packet is reflected rather than transmitted through the slits.

The real and imaginary parts of the wavefunction, plotted in Figures 3–4, behave consistently with the analytic form of the initial condition. The initial packet shows the expected oscillatory structure in x due to the imposed momentum $p_x = 200$, while remaining slowly varying in y. Writing

$$u(x, y, 0) = C(x, y)\, e^{i\theta(x,y)} = C(x, y)\left[\cos\theta(x, y) + i \sin\theta(x, y)\right],$$

we expect $\mathrm{Re}(u)$ to follow a cosine pattern and $\mathrm{Im}(u)$ a sine pattern, with an approximate phase shift of $\pi/2$ between the two components[2]. This relationship is visible in all three time slices. After the packet encounters the barrier, both $\mathrm{Re}(u)$ and $\mathrm{Im}(u)$ develop curved wavefronts emerging from the slit openings, and their oscillatory structure becomes more complex as the transmitted parts from the two slits interfere. The resulting phase variations are consistent with the emerging interference fringes observed in the probability density.

Comparative Interference Patterns from Single-, Double-, and Triple-Slit Configurations

We extended our simulation framework to investigate detection probabilities for single-, double-, and triple-slit configurations. Specifically, we measured the particle distribution along a detector screen placed at $x = 0.8$ at time $t = 0.002$, thereby extracting the one-dimensional probability function $p(y \mid x = 0.8; t = 0.002)$. The numerical setup was kept consistent with the parameters in Table II; only the number of slits was varied. Each slit has width 0.05 and neighbouring slits are separated by barriers of length 0.05.

Operationally, $p(y \mid x = 0.8; t = 0.002)$ was obtained by sampling the discrete wave function $u(x_i, y_j, t_n)$ along the grid line closest to $x = 0.8$ at the time index corresponding to $t = 0.002$. We computed $q_j =$

[2] A detailed derivation is given in Appendix B.

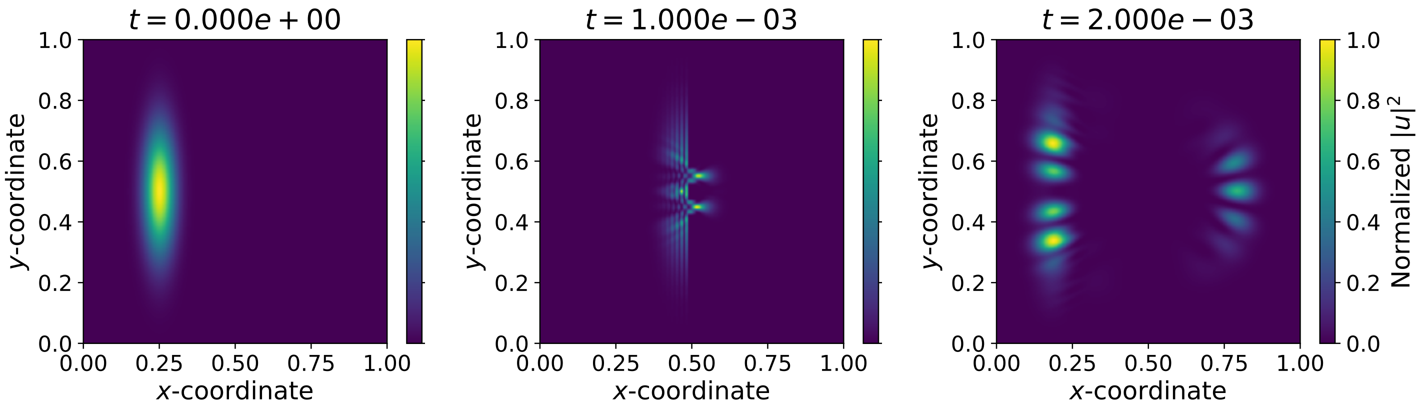

FIG. 2: Time evolution of the probability distribution $|u(x,y,t)|^2$ at $t = 0$, $t = 0.001$, and $t = 0.002$. At $t = 0$, the wave packet is a localised Gaussian centred at $x = 0.25$. By $t = 0.001$, only the components aligned with the two slit openings transmit through the barrier. At $t = 0.002$, the parts of the wave that pass through the two slits start to overlap and create an interference pattern, while most of the wave is still stuck on the left side because it bounced off the barrier.

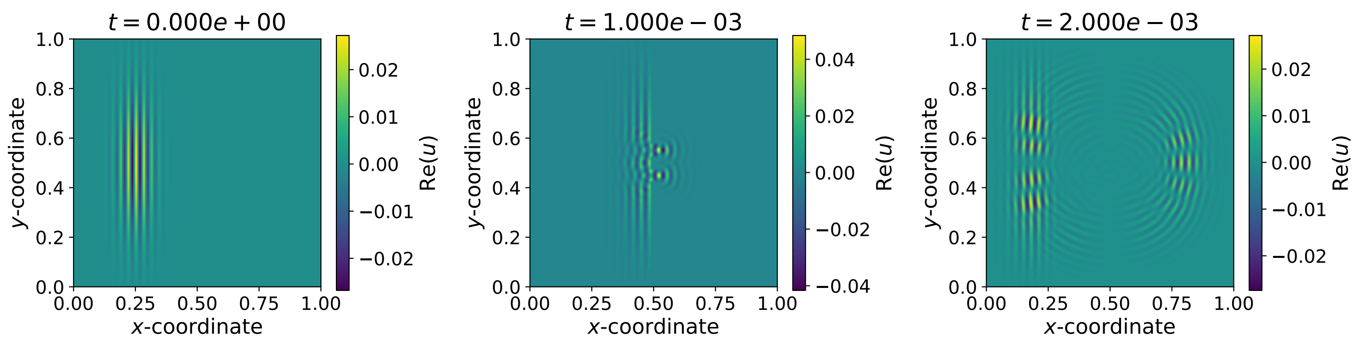

FIG. 3: Real part of the wave function $\mathrm{Re}\big(u(x,y,t)\big)$ at $t = 0$, $t = 0.001$, and $t = 0.002$. The initial state shows a vertically elongated Gaussian enveloped around $x \approx 0.25$, with clear oscillations along the x-direction and essentially a single peak in y. After the packet reaches the double slit, two outgoing wavefronts emerge from the slit openings, and by $t = 0.002$ the overlapping wavefronts form an interference pattern consistent with double-slit diffraction.

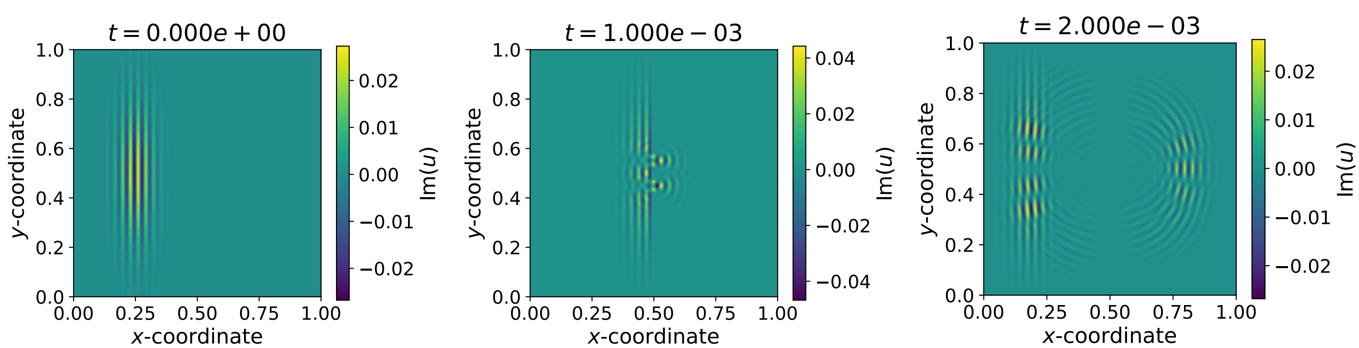

FIG. 4: Imaginary part of the wave function $\mathrm{Im}\big(u(x,y,t)\big)$ at $t = 0$, $t = 0.001$, and $t = 0.002$. The initial state shows the expected sine-modulated Gaussian profile. After encountering the double slit, two outgoing wavefronts emerge from the slit openings, and by $t = 0.002$ their overlap produces an interference pattern analogous to that seen in the real part.

$|u(x_{\text{screen}}, y_j, t = 0.002)|^2$ and normalised along y so that $\sum_j p_j = 1$, with $p_j = q_j / \sum_k q_k$.

The resulting detection profiles are shown in Fig. 5. All three profiles are approximately symmetric around $y = 0.5$, and the number of visible peaks increases from one (single slit) to several fringes for the double and triple slits. For the single slit we observe a broad central peak that dissipates on the sides. In the double-slit case the distribution is modulated by clearly visible fringes with an approximately regular spacing. The triple-slit profile shows a more structured pattern, with narrower main peaks and additional smaller maxima. These features are consistent with the qualitative expectations for diffraction and interference from one, two and multiple slits.

Note that the simulation is terminated before boundary reflections can perturb the observed pattern.

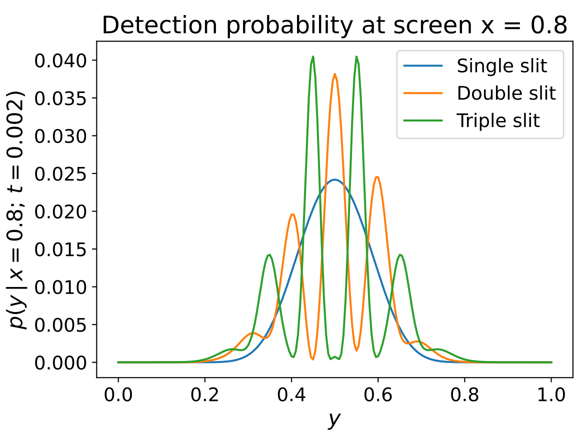

FIG. 5: Conditional detection probability $p(y \mid x = 0.8; t = 0.002)$ along the vertical detector screen for single-, double-, and triple-slit configurations. All simulations use identical numerical parameters, and only the number of slits is changed. The single slit gives a broad central peak, while the double and triple slits produce increasingly structured interference patterns.

IV. CONCLUSION

In this work, we implemented a Crank–Nicolson solver for the 2D Schrödinger equation and used it to simulate Gaussian wave-packet propagation through various slit potentials.

The Crank–Nicolson solver conserves total probability to near machine precision in both free propagation and double-slit simulations. Throughout the simulation interval, the deviation $|P(t) - 1|$ remains below 10^{-13} without systematic drift, confirming the method's norm preservation and the stability afforded by the direct LU solver. Probability conservation is similarly unaffected by the sharply varying double-slit potential, indicating robustness even in the presence of discontinuous barriers. A broader study varying h, Δt, potential strength, and total simulation time would be required to fully map the scheme's stability limits. Overall, the strong probability conservation validates the correctness and reliability of the numerical implementation.

The simulated Gaussian wave packet displays the expected behaviour: predominant reflection from the barrier and two transmitted components that form curved wavefronts and the early stages of an interference pattern. Only a small fraction of the wave passes through the slits, leaving most of the probability on the left side of the domain. The real and imaginary parts evolve consistently with the initial phase structure and with the interference responsible for the modulation in $|u|^2$. Small deviations in total probability stem from finite spatial and temporal resolution. Because of the short final time and chosen grid spacing, the interference fringes are not fully developed; extending the simulation time, refining the grid, or enlarging the domain would allow clearer resolution of the pattern.

Detection probabilities $p(y \mid x = 0.8, t = 0.002)$ for single-, double-, and triple-slit geometries show the expected progression of diffraction and interference complexity: a broad single-slit peak, well-defined double-slit fringes, and a more intricate triple-slit pattern with finer structure. These results align with qualitative expectations for multi-slit interference. Further work could systematically explore how the profiles depend on slit spacing, aperture width, and detector position, as well as compare the numerical distributions with analytical diffraction models.

Appendix A: Crank–Nicolson Discretisation of the 2D Schrödinger Equation

Starting from the scaled Schrödinger equation (1)

$$i\frac{\partial u}{\partial t} = -\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} + v(x,y)u,$$

we rewrite it as

$$\frac{\partial u}{\partial t} = i\left(u_{xx} + u_{yy} - vu\right) = i\left(\Delta_2 u - vu\right).$$

Crank–Nicolson for an equation of the form $u_t = F(u)$ is

$$\frac{u^{n+1} - u^n}{\Delta t} = \frac{1}{2}\left[F(u^{n+1}) + F(u^n)\right].$$

With $F(u) = i(\Delta_2 u - vu)$, we obtain at grid point (x_i, y_j):

$$\frac{u_{ij}^{n+1} - u_{ij}^n}{\Delta t} = \frac{i}{2}\left[\Delta_2 u_{ij}^{n+1} - v_{ij}u_{ij}^{n+1} + \Delta_2 u_{ij}^n - v_{ij}u_{ij}^n\right].$$

Multiplying by Δt:

$$u_{ij}^{n+1} - u_{ij}^n = \frac{i\Delta t}{2}\left[\Delta_2 u_{ij}^{n+1} - v_{ij}u_{ij}^{n+1} + \Delta_2 u_{ij}^n - v_{ij}u_{ij}^n\right].$$

Bringing all $n+1$ terms to the left:

$$u_{ij}^{n+1}-\frac{i\Delta t}{2}\left[\Delta_2 u_{ij}^{n+1}-v_{ij}u_{ij}^{n+1}\right]=u_{ij}^{n}+\frac{i\Delta t}{2}\left[\Delta_2 u_{ij}^{n}-v_{ij}u_{ij}^{n}\right].$$

We now discretize the Laplacian using centered finite differences:

$$\Delta_2 u_{ij}^n \approx \frac{u_{i+1,j}^n-2u_{ij}^n+u_{i-1,j}^n+u_{i,j+1}^n-2u_{ij}^n+u_{i,j-1}^n}{h^2}.$$

Define

$$r=\frac{i\Delta t}{2h^2}.$$

Substituting the discrete Laplacian into the CN equation yields:

$$\begin{aligned}
&u_{ij}^{n+1}-r\big(u_{i+1,j}^{n+1}-2u_{ij}^{n+1}+u_{i-1,j}^{n+1}\big)\\
&\quad-r\big(u_{i,j+1}^{n+1}-2u_{ij}^{n+1}+u_{i,j-1}^{n+1}\big)+\frac{\mathrm{i}\Delta t}{2}v_{ij}u_{ij}^{n+1}\\
&\quad=\\
&\quad u_{ij}^{n}+r\big(u_{i+1,j}^{n}-2u_{ij}^{n}+u_{i-1,j}^{n}\big)\\
&\qquad+r\big(u_{i,j+1}^{n}-2u_{ij}^{n}+u_{i,j-1}^{n}\big)-\frac{\mathrm{i}\Delta t}{2}v_{ij}u_{ij}^{n},
\end{aligned}$$

This is the Crank–Nicolson update equation for the 2-D Schrödinger equation [11].

Appendix B: Amplitude–Phase Decomposition of the Initial Wave Packet

In order to rewrite equation (11)

$$u(x,y,0)=\exp\left[-\frac{(x-x_c)^2}{2\sigma_x^2}-\frac{(y-y_c)^2}{2\sigma_y^2}+ip_xx+ip_yy\right],$$

in terms of an amplitude and a phase, we separate the real and imaginary parts of the exponent:

$$u(x,y,0)=\exp\left[-\frac{(x-x_c)^2}{2\sigma_x^2}-\frac{(y-y_c)^2}{2\sigma_y^2}\right]\exp[i\,(p_xx+p_yy)].$$

Then we define

$$C(x,y)=\exp\left[-\frac{(x-x_c)^2}{2\sigma_x^2}-\frac{(y-y_c)^2}{2\sigma_y^2}\right],$$

and

$$\theta(x,y)=p_xx+p_yy,$$

which gives us

$$u(x,y,0)=C(x,y)\,e^{i\theta(x,y)}.$$

Using Euler's formula $e^{i\theta}=\cos\theta+i\sin\theta$ yields

$$u(x,y,0)=C(x,y)\big[\cos\theta(x,y)+i\sin\theta(x,y)\big].$$

From this representation,

$$\mathrm{Re}(u)=C(x,y)\cos\theta(x,y),\qquad \mathrm{Im}(u)=C(x,y)\sin\theta(x,y),$$

so the real and imaginary parts differ by a phase shift of $\pi/2$, as observed in the numerical results.

[1] David J. Griffiths and Darrell F. Schroeter. *Introduction to Quantum Mechanics*. Cambridge University Press, third edition, 2018.

[2] Hui Peng and James Peng. Experimental study of mystery of double slit — comprehensive double slit experiments. *International Journal of Physics*, 9:114–127, 03 2021.

[3] Anders Kvellestad. FYS3150 - Lecture Notes. https://github.com/anderkve/FYS3150/blob/master/lecture_notes/2025/lecture_notes.pdf, 2025. Version 1.18.3, page 195–197. Accessed on November 25, 2025.

[4] Anders Kvellestad. The Schrödinger equation. https://anderkve.github.io/FYS3150/book/projects/project5.html#the-schrodinger-equation, 2025. Accessed on November 23, 2025.

[5] Anders Kvellestad. Initial and boundary conditions. https://anderkve.github.io/FYS3150/book/projects/project5.html#initial-and-boundary-conditions, 2025. Accessed on November 24, 2025.

[6] Anders Kvellestad. FYS3150 - Lecture Notes. https://github.com/anderkve/FYS3150/blob/master/lecture_notes/2025/lecture_notes.pdf, 2025. Version 1.18.3, page 15. Accessed on November 26, 2025.

[7] Anders Kvellestad. FYS3150 - Lecture Notes. https://github.com/anderkve/FYS3150/blob/master/lecture_notes/2025/lecture_notes.pdf, 2025. Version 1.18.3, page 187-188. Accessed on November 26, 2025.

[8] Anders Kvellestad. FYS3150 - Lecture Notes. https://github.com/anderkve/FYS3150/blob/master/lecture_notes/2025/lecture_notes.pdf, 2025. Version 1.18.3, page 195-197. Accessed on November 26, 2025.

[9] Ebenezer Bonyah Rashid Jan Muhammad Nisar Abdel-Haleem Abdel-Aty Ibrahim S. Yahia Amin Khan, Muhammad Ahsan. Numerical solution of schrödinger equation by crank–nicolson method. https://onlinelibrary.wiley.com/doi/10.1155/2022/6991067, 2022. Discusses stability and second-order accuracy of the Crank–Nicolson scheme for the Schrödinger equation. Accessed on December 2, 2025.

[10] Etienne Thibierge Loren Jørgensen, David Lopes Cardozo. Numerical resolution of the schrödinger equation. https://web.pa.msu.edu/people/duxbury/courses/phy480/SchrodingerDynamics.pdf, 2011. Section 2.3.2 on Accuracy. Accessed on December 4, 2025.

[11] Anders Kvellestad. FYS3150 - Lecture Notes. https://github.com/anderkve/FYS3150/blob/master/lecture_notes/2025/lecture_notes.pdf, 2025. Version 1.18.3, page 194-201. Accessed on December 8, 2025.

[12] Anders Kvellestad. FYS3150 - Lecture Notes. https://github.com/anderkve/FYS3150/blob/master/lecture_notes/2025/lecture_notes.pdf, 2025. Version 1.18.3, page 198-201. Accessed on November 27, 2025.